Samuel R Bray, PhD candidate

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RESEARCH INTERESTS

- Emergence of complex function from ensembles of biological units
- Generating and leveraging large-scale dynamics data to understand latent biological systems
- Interpretable machine learning for synthesizing massive data

EDUCATION

Stanford University (2016-November 2022)
Advisor: Bo Wang
Ph.D. Bioengineering (*candidate*)
M.S. Bioengineering
GPA: 3.966

UCLA (2012-2016)
B.S. Bioengineering
Technical Breadth, Computer Science
GPA: 3.925

SKILL SETS

Computation

- Machine learning
- Statistical mechanics
- Python / C++ / Matlab
- Keras / Tensorflow / Pytorch
- Time series modeling + analysis
- Image segmentation + analysis
- Database design + management
- Single cell RNA sequencing analysis

Biology

- Neuroscience
- Behavior science
- Developmental biology
- Ecology
- Microscopy
- Cloning / PCR / transfection
- Microfluidics
- Cell culture
- Fabrication (laser cutter, soldering, etc.)

RESEARCH EXPERIENCE

Differential robustness of neural networks in a regenerating brain — 2019-present
Supervisor: Bo Wang, Stanford

- Developed imaging platform and video analysis pipeline to screen for behavioral variation and phenotypes in planarian flatworms during neural regeneration and gene knockdown (*github:* **planameterization**).
- Used available single-cell RNAseq atlas to map differentiation trajectories from stem cells to diverse neural subtypes in planarian (**Wyss et al, 2022**, github: **neuralRegeneration_review**)
- Built and analyzed database of 60+ genetic knockdowns (>180 TB of video data) under a variety of custom-built stimulus modalities and protocols.
- Wrote easy-to-use repository for simple data exploration for non-experts. (*github**: behavior-lite***)
- Designed and trained custom Recurrent Neural Network (RNN) architecture to study the interplay between volumetric neuropeptide signaling and synaptic transmission in neural dynamics (*github:* **dualChannelRNN**)

Inferring latent dynamics in time series data — 2020-present
Supervisor: Bo Wang, Stanford

- Designed novel machine-learning architecture to parameterize latent models from partially-observed experimental data (github**: ParEO**)
- Used Variational Autoencoding (VAE) of video segments and generative time-series modeling to visualize the space of planarian movements and generate synthetic videos of their behavior responses (github: **thisWormDNE**)

Statistical mechanics in ecology 2017-2020

Supervisor: Bo Wang, Stanford

- Designed *in silico* and *in vitro* bacteria communities to explore phase transitions in ecosystems ***(Bray et. al, 2019)***
- Measured novel signatures of criticality decade+ long ecological data. Designed a method to use this information to improve time-series forecasting *(**Bray, Wang. 2020,** github: **ecologicalAvalanches**)*
- Assembled auxotrophic E.coli communities to measure small-scale community fluctuations using microfluidics and expansion microscopy *(**Lim et. al, 2019**)*

E. coli whole cell model 2017

Supervisor: Markus Covert, Stanford

- Studied transgenerational effects of transcriptional bursts in a computational model of E.coli **(Macklin et. al, 2020)**

PUBLICATIONS (* denotes equal contribution)

Bray S.R.*, Wyss L.*, Wang B. "Adaptive robustness through incoherent signaling mechanisms in a regenerative brain" (*submitted*)

Wyss L.*, **Bray S.R.***, Wang B. "Cellular diversity and plasticity of planarian nervous system" Curr. Opin. Genet. Dev. 2022

Bray S.R., Wang B. "Forecasting unprecedented critical events in ecological fluctuations" PLOS Comp. Bio. 2020

Macklin D., … **Bray S.R.** … Covert M. "Simultaneous cross-evaluation of heterogeneous E. coli datasets via mechanistic simulation" Science. 2020

Bray S.R., Fan Y., Wang B. "Phase transitions in mutualistic communities under invasion" Physical Biology. 2019

Lim Y, … **Bray S.R.** … Wang B. "Mechanically resolved imaging of bacteria using expansion microscopy" PLOS Biology. 2019

PRESENTATIONS

Bray S.R.*, Wyss L.*, Wang B. "Differential robustness of neural networks in a regenerative brain" APS March Meeting. 2022

Bray S.R., Wang B. "Universal avalanche fluctuations in ecological communities under resource frustration" APS March Meeting. 2019

Bray S.R., Fan Y., Wang B. "Invasion-induced phase transitions in microbial ecosystems" APS March Meeting. 2018

Karimi A., Roy R., **Bray S.R.**, Di Carlo D., "Controlling Lateral Inertial Migration Rate of Particles in Microchannels" APS/DFD. 2015

PUBLIC GITHUB PROJECTS

Planarian behavior database github.com/samuelbray32/behavior-lite

- Repository to easily explore database of planarian knockdown and regeneration phenotypes
- Friendly interface for execution of numerous measurements and bootstrap statistical testing

Automated behavior parameterization github.com/samuelbray32/planameterization

- Image segmentation and parameterization
- Labeling of continuous (LDS) or discrete (HMM) latent states
- One-line fit and transformation functions

Dual-channel neural networks github.com/samuelbray32/dualChannelRNN

- Mechanistic model of mixed diffusive and synaptic neural signaling with a keras implementation
- Trainable to simultaneously match animal behaviors under a variety of genetic knockdowns
- Efficient simulation of neural ablation to study network robustness

Latent model parameterization github.com/samuelbray32/ParEO

- Parameter Estimate from Observables (ParEO)
- Learned latent space reconstruction and mechanistic model fitting using the keras framework
- Applicable to non-linear and chaotic systems without analytical solutions or known initial conditions

Single cell neuron development github.com/samuelbray32/neuralRegeneration_review

- Single cell sequencing analysis of planarian neural atlas
- Analysis of changing transcriptional coregulation during neuron differentiation and maturation

Forecasting dynamics in ecology github.com/samuelbray32/ecologicalAvalanches

- Analysis module for fitting avalanche statistics
- Module for empirical dynamic forecasting
- Integration of Avalanche Scaling Extrapolation (ASE)

Non-stationary Markov models github.com/samuelbray32/variable_hmm

- Modifies C-level message passing in sci-kit HMM to include time-dependent transition matrix
- Optimizes time-dependent parameters using explicit likelihood derivatives

This worm does not exist github.com/samuelbray32/thisWormDNE

- Compresses video clips of movement using VAE
- Fits stochastic model to embedded timeseries
- Generates synthetic videos of behavioral response

TEACHING + MENTORSHIP

Teaching assistant, Stanford

• Fundamentals for Engineering Biology Lab (2018, 2019, 2022)
Supervisor: Zelda Love
Responsibilities: Training of undergraduates in synthetic biology techniques. Supervision of student groups in design, production, and testing of novel gene constructs.

• Systems Biology (2017)
Supervisor: Markus Covert
Responsibilities: Preparation and delivery of weekly review lectures. Design of homework, midterm, and final exams. Hosted weekly office hours.

Undergraduate/High school research mentor

Jacob Qian (2022)
Maria Lozada (2019)
Xochitl Longstaff (2018)
Fan Yuhang (2017)
Stella Wan (2017)